JENNIFER ZHU

ACADEMIC CURRICULUM VITAE

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RESEARCH PROFILE

Biomedical researcher trained in health data science and statistics, working across quantitative MRI, radiopathomic analysis, longitudinal oncology data, statistical modeling, and applied machine learning. Research interests include computational imaging, biomedical ML, and computational neuroscience.

EDUCATION

M.S., Health Data Science | University of California, San Francisco | June 2026

B.S., Statistics | San Francisco State University | December 2022

RESEARCH EXPERIENCE

Graduate Student Researcher | UCSF, Lupo Lab | June 2025–Present

- Analyze longitudinal multiparametric MRI and clinical outcomes across 95 patients with newly diagnosed glioblastoma from three research cohorts, covering pre- and post-radiotherapy, pre-progression, and progression scans.
- Design and adapt ROI-based quantitative imaging feature workflows for contrast-enhancing and non-enhancing lesions, including distributional statistics, tumor-volume changes, and radiopathomic cellularity and tumor-probability maps.
- Quantify threshold-defined abnormal regions using intensity thresholds, morphological cleaning, connected-component filtering, volumetric features, and imaging quality control.
- Perform Kaplan–Meier, multivariable Cox regression, linear regression, model diagnostics, and false discovery rate adjustment to assess imaging–outcome associations.
- Create MRI overlays, survival plots, manuscript figures, and scientific summaries for multidisciplinary collaborators; first-author manuscript in preparation.

MANUSCRIPT IN PREPARATION

First-author manuscript in preparation on longitudinal radiopathomic biomarkers in glioblastoma. Title and publication status to be finalized before public submission; results not publicly disclosed.

COMPUTATIONAL & STATISTICAL RESEARCH PROJECTS

BloodMNIST — Biomedical Computer Vision

- Collaborative project comparing a PyTorch convolutional neural network with a random-forest baseline on eight classes of microscopy images.
- CNN achieved 90.76% held-out test accuracy; evaluated precision, recall, F1 and ROC-AUC across classes.

METABRIC — Clinical Survival Analysis

- Analyzed hormone therapy and recurrence-free survival in an ER-positive breast cancer cohort using Kaplan–Meier, multiple imputation, and adjusted/stratified Cox regression.
- Used Schoenfeld residuals for proportional-hazards diagnostics and Rubin's rules for pooling estimates.

I-SPY2 — Bayesian Modeling of Treatment Response

- Fit Bayesian logistic regression models of pathological complete response using baseline MRI tumor diameter, sphericity and HR/HER2 status.
- Compared three models using leave-one-out cross-validation; assessed posterior distributions, MCMC convergence and prior sensitivity on 384 complete cases.

Breast Tumor Classification — PCA & QDA

- Applied PCA and quadratic discriminant analysis to Wisconsin breast cancer cell-nucleus features.
- Reported 93.9% test accuracy and 0.991 ROC-AUC; assessed class-specific recall and false-negative risk.

ADDITIONAL PROJECT

Brain Tumor Clinical Feature Classification — Compared logistic regression, random forest and XGBoost on a structured benchmark; evaluated feature importance, reduced-feature models, sensitivity and specificity.

TECHNICAL METHODS & SOFTWARE

Programming: Python, R, SQL, MATLAB, SAS; pandas, NumPy, tidyverse, Jupyter, R Markdown, Git, Linux

Statistical Methods: Hypothesis testing, ANOVA, categorical data analysis, linear/logistic regression, multivariable models, Cox proportional hazards, Kaplan–Meier, Schoenfeld residuals, stratification, MICE, Rubin's rules, Bayesian inference, MCMC, posterior predictive checks, LOO-CV, FDR

Machine Learning: PyTorch, scikit-learn, CNNs, random forests, XGBoost, PCA, QDA, feature engineering, model validation, ROC-AUC/F1/precision/recall

Imaging & Visualization: Quantitative MRI, radiopathomic feature extraction, NIfTI, segmentation QC, morphology, Matplotlib, ggplot2, scientific figures, survival plots, image overlays

Research Interests: Computational neuroscience, EEG/ECoG, neural decoding and BCI (in development)